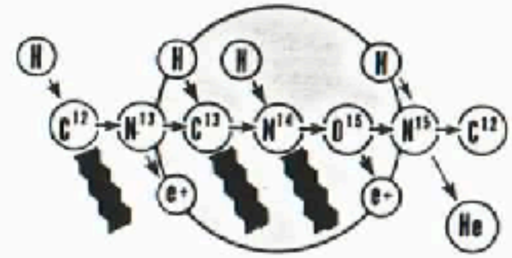
PERIODIC TABLE AND ATOMIC DATA

WITH ILLUSTRATED TEXT OF NUCLEAR TERMS

The Source of Solar Energy

Illustrating the solar carbon-nitrogen cycle which results in the fusion of four hydrogen nuclei into one helium nucleus. A hydrogen nucleus unites with a carbon nucleus (C12) changing it to an isotope of nitrogen (N^{13}) and releasing large amounts of energy. The unstable N^{13} releases a positive electron (e+) and changes to stable C13. C^{13}, penetrated by another hydrogen nucleus, changes to N^{14} with much energy release. The cycle continues until finally the fourth hydrogen nucleus causes N^{15} to be converted to the original C^{12} with the creation of a helium nucleus.

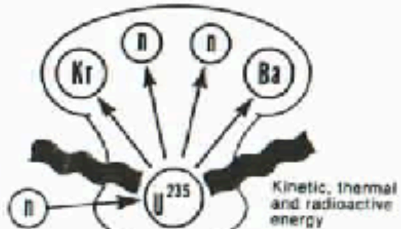


	IA	IIA	IIIB	IVB	VB	VIB	VIIB	VIII	VIII	VIII	IB	IIB	IIIA	IVA	VA	VIA	VIIA	INERT GASES
1	1 H Hydrogen 1.008																	2 He Helium 4.003
2	3 Li Lithium 6.939	4 Be Beryllium 9.012											5 B Boron 10.81	6 C Carbon 12.01	7 N Nitrogen 14.01	8 O Oxygen 15.999	9 F Fluorine 19.00	10 Ne Neon 20.18
3	11 Na Sodium 22.99	12 Mg Magnesium 24.31											13 Al Aluminum 26.98	14 Si Silicon 28.09	15 P Phosphorus 30.97	16 S Sulphur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.95
4	19 K Potassium 39.10	20 Ca Calcium 40.08	21 Sc Scandium 44.96	22 Ti Titanium 47.90	23 V Vanadium 50.94	24 Cr Chromium 52.00	25 Mn Manganese 54.94	26 Fe Iron 55.85	27 Co Cobalt 58.93	28 Ni Nickel 58.71	29 Cu Copper 63.54	30 Zn Zinc 65.37	31 Ga Gallium 69.72	32 Ge Germanium 72.59	33 As Arsenic 74.92	34 Se Selenium 78.96	35 Br Bromine 79.91	36 Kr Krypton 83.80
5	37 Rb Rubidium 85.47	38 Sr Strontium 87.62	39 Y Yttrium 88.91	40 Zr Zirconium 91.22	41 Nb Niobium 92.91	42 Mo Molybdenum 95.94	43 Tc Technetium (99)	44 Ru Ruthenium 101.1	45 Rh Rhodium 102.9	46 Pd Palladium 106.4	47 Ag Silver 107.9	48 Cd Cadmium 112.4	49 In Indium 114.8	50 Sn Tin 118.7	51 Sb Antimony 121.8	52 Te Tellurium 127.6	53 I Iodine 126.9	54 Xe Xenon 131.3
6	55 Cs Cesium 132.9	56 Ba Barium 137.3		72 Hf Hafnium 178.5	73 Ta Tantalum 180.9	74 W Tungsten 183.9	75 Re Rhenium 186.2	76 Os Osmium 190.2	77 Ir Iridium 192.2	78 Pt Platinum 195.1	79 Au Gold 197.0	80 Hg Mercury 200.6	81 Tl Thallium 204.4	82 Pb Lead 207.2	83 Bi Bismuth 209.0	84 Po Polonium (210)	85 At Astatine (210)	86 Rn Radon (222)
7	87 Fr Francium (223)	88 Ra Radium (226)																

57 La Lanthanum 138.9	58 Ce Cerium 140.1	59 Pr Praseodymium 140.9	60 Nd Neodymium 144.2	61 Pm Promethium (147)	62 Sm Samarium 150.4	63 Eu Europium 152.0	64 Gd Gadolinium 157.3	65 Tb Terbium 158.9	66 Dy Dysprosium 162.5	67 Ho Holmium 164.9	68 Er Erbium 167.3	69 Tm Thulium 168.9	70 Yb Ytterbium 173.0	71 Lu Lutetium 175.0
89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium (231)	92 U Uranium 238.0	93 Np Neptunium (237)	94 Pu Plutonium (242)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (249)	98 Cf Californium (251)	99 Es Einsteinium (254)	100 Fm Fermium (253)	101 Md Mendelevium (256)	102 No Nobelium (255)	103 Lr Lawrencium (257)

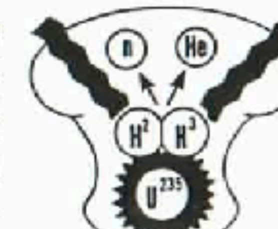
Atomic Fission Process

This simplified diagram illustrates a form of nuclear fission used in the original atom bomb. A neutron (n) is forced to penetrate an atom of uranium 235. The uranium nucleus breaks up into nuclei of krypton and barium with release of 2 neutrons and large quantities of energy. Each released neutron then penetrates another uranium atom to cause an explosive chain reaction.



Thermonuclear Fusion Process

A form of hydrogen reaction used in military weapons is illustrated by this diagram. A source of intense heat, shown here as a uranium fission device, causes the fusion of two hydrogen isotopes — deuterium (H2) and tritium (H3). Helium nuclei are thus formed along with the release of neutrons and tremendous energy. No chain reaction takes place.



Elements 104 and 105

Existance confirmed but names and properties still under consideration. 104 and 105 start the transactinide series and are members of the 6d transition series. Tentative names for 104 (located under Hafnium) are Rutherfordium (U.S.A.), Kurchatovium (U.S.S.R.). Atomic weight 261, half-life 60 sec. Tentative names for 105 (located under Tantalum) are Hahnium (U.S.A.), Bohrium (U.S.S.R.). Atomic weight 262, half-life 40 sec. Chemical properties of each are similar to IV B and V B groups respectively.

GLOSSARY OF N CLEAR TERMS

The chemical and physical properties of the elements are periodic functions of their atomic structure. Therefore, atoms with similar structure have properties which repeat at regular intervals.

A periodic table shows the similarity between elements at regular intervals and thus permits systematic study of their properties.

Dmitri Mendeleev in 1869 created the first periodic arrangement that satisfactorily accounted for gaps in previous tables. His arrangement is the basis for all modern periodic tables.

Lothar Meyer, working at the same time, first presented graphs showing the periodic nature of various properties of the elements. His work is the basis for modern graphs showing periodic functions.

In a periodic table the number of the horizontal period (1, 2, 3, 4, 5, 6, or 7) is identical to the number of orbits in any element shown in that period. The only exception is No. 46, palladium.

The elements in each vertical group resemble each other closely in various properties. Also, elements in the A groups have characteristics similar to B group elements of the same column number.

As an example of group similarity, note that copper, silver, and gold (group IB) are all heavy metals, are not highly reactive, are found in a relatively free state and each contains one electron in its outer orbit.

The inert gases (column 0) are a striking example of periodic similarity in that none of them unite with other elements to form compounds. This is caused by each having all of its electron orbits filled to capacity.

Most of the vertical groups show impressive similarities between their elements. Reference to a basic chemistry manual will explain these properties.

The rare earth elements (57 to 71) and the uranium metal series (89 to 103) have striking similarities within their own groups. Further study of these unusual elements is suggested.

Elementary particles are the building blocks of all matter. True elementary particles cannot be broken into smaller units and have no internal structure.

Subatomic particles are components of atoms. They are classified into three families — photons, hadrons and leptons — according to their mode of interaction.

A photon is a particle or quantum of light energy. It is the only particle which interacts electromagnetically. It is postulated as the carrier of electromagnetic force.

Hadrons are a class of subatomic particles which take part in strong nuclear interactions. Baryons and mesons are subclasses of particles within the hadron family.

A quark is the constituent particle of hadrons. It is a true elementary particle.

The gluon is postulated as a carrying particle for the strong nuclear force which binds or "glues" quarks together.

A proton is a positively charged subatomic particle found in the nucleus of an atom. It is a member of the hadron family and is composed of quarks.

A neutron is an uncharged subatomic particle found in the nucleus of the atom. It is a member of the hadron family and is composed of quarks.

Leptons are a class of subatomic particles which take part in weak nuclear interactions.

The boson is postulated as a carrying particle for the weak nuclear force associated with leptons.

An electron is a negatively charged subatomic particle which orbits around the nucleus of an atom. It is a member of the lepton family. Electrons, unlike protons, are true elementary particles.

An atom is the smallest identifiable unit of a given type of matter. Further division would destroy the atom and convert it to energy and subatomic particles.

A molecule is the smallest identifiable unit of a given substance. Further division would destroy the identity of the substance by converting it into its individual atoms.

An element is a substance composed of one type of atom.

A compound is a substance composed of two or more types of atoms.

A molecule of an element is usually made up of one atom of that element, but in some cases—such as gaseous elements—a molecule may contain more than one identical atom.

An atom is a miniature planetary system consisting of a nucleus or center around which revolves one or more electrons.

Atomic number is the number of protons in the nucleus of an atom. Atomic number can also be regarded as the magnitude of the positive charge in the nucleus.

The total number of electrons in the orbits of an atom is equal to the number of protons in the nucleus. Therefore, the negative electron charge is equal to the positive proton charge. See the atomic structure diagram on this chart.

A nucleon is either a neutron or proton. The term refers to a component of the nucleus.

Atomic weight is the relative weight of an atom when a specific isotope of the carbon atom is assigned a value of 12.

A single atom of a light element may weigh more than an atom of a heavier element. One example is the atomic weight of oxygen and carbon. The atoms of the light element may be spaced more loosely than in the heavier element.

Mass number is the total number of protons and neutrons in the nucleus of an atom. The atomic weight of an element is influenced by the number of neutrons in its nucleus. See isotope.

An ion is an atom that has lost or acquired an electron, thus changing it from a neutral to an either positive or negative state.

An anion is a negative ion—an extra electron is present.

Ionization potential is the energy required to remove an electron from an atom, thus forming a positive ion.

Valence number is the number of electrons that an element will lose, gain or share when it forms a compound.

Positive valence exists when electrons are lost; negative valence exists when electrons are gained.

The valence orbit is the outermost orbit of an atom. This orbit largely determines the chemical properties of an element. Similar outer orbits denote similar chemical characteristics. The inner orbit of an atom never contains more than 2 electrons; the outer orbit never contains more than 8 electrons. No. 46, palladium, is the only exception.

An isotope is a variation of an atom or element that is chemically identical but with a different number of neutrons in its nucleus. Therefore, an isotope has a different atomic weight and mass number than its counterpart.

Some elements have no natural variations or isotopes. Other elements may have from two to ten natural isotopes.

A radioactive isotope is an atom that constantly emits nuclear energy. Such isotopes are found naturally but more often are created in nuclear laboratories.

Half-life is a measure of the rate of disintegration of a radioactive substance. Half-life is the time required for a substance to give up one-half of its radioactive energy.

Nuclear fission is the process of disrupting the nucleus of an atom so that different nuclei are formed along with the release of great quantities of energy.

Thermonuclear fusion is the process of uniting two or more nuclei to create a different nucleus with subsequent tremendous release of energy.

Solar energy is created by thermonuclear fusion. Broadly stated, four hydrogen nuclei unite to form one helium nucleus with release of radiant energy. See front of chart.

Alpha rays are streams of helium nuclei emitted from radioactive substances. An alpha particle is a helium atom without its two electrons; thus it bears a double positive charge.

Beta rays are streams of electrons emitted during radioactive decomposition. A beta particle has a negative charge and has 1/1845 the mass of an alpha particle.

Gamma rays are high frequency energy radiations released during radioactive processes. Gamma rays are similar to X-rays and fall between them and cosmic rays on the radiant spectrum.

Radioactivity varies in intensity. Alpha rays can be shielded with paper but heavy lead foil is required for beta rays. The dangerous gamma rays require extensive shielding precautions.

Einstein's theory of relativity is closely related to the mass-energy concept of the atom. One teaching of the theory states that the energy that can be released from a given quantity of matter is equivalent to its mass multiplied by the speed of light squared. The well known $E=mc^2$ equation.

Thermo Gold? To a p pts powder dusting, heating to melting point of Silver less 10% of that temperature and drop, donot enclose.

Sand? Will convert to mercury.

Mercury? It converts to crystal in carbon.

Tin? Makes platinum

[illegible] Bubble? The metal is now a alloy of copper and because of the magnetic properties of that compound it changed the specifics of the container and its surrounding silica, the alloy is of copper and silicon.

[illegible] alloy? has never been created before and is a Conductive Silicate.

[illegible] Cubes? a p p of dust to copper, use a geometric shape for piece of copper and stay within one half to one thousandth of one meter.

names? "Ranusartiniums"

process? "Chemoradification" — a Combination of the Rare earth elements.

Ruby Material? (Is it Ruby?) It is crystal Frequency of Copper attributes and its Ruby, it was Caused by magnetochemical Callyst when vibrating wave lengths of the callyst and the Copper reached a threshold when excited by heating.

The Red Powder? It was ferric extract and a unusual combination of inert nuclei and its formulation is unstable when heated.

[illegible]
Refining.?? In its constituted attributes it of course can be further refined, much experimenting must be done. the wave problem being the control factor in the extraction of the rare materials

Rare Materials? Here are combinations with the Rare[illegible] and inert gases held in suspension.

...Nuggets From Little Bacteria Grow, Chemis...

By Bryan Bashin
Special to The Bee

For decades, scientists have wrestled with the puzzle of how placer gold — the flakes and nuggets that lie mixed with river sand in gold-rich country like Northern California — also can be formed in areas where only low-grade gold ore is present. Now, in an unexpected discovery, they've found that bacteria form the nuggets.

According to U.S. Geological Survey research chemist John Watterson, the bacillus cereus bacteria actually crystallizes dissolved gold out of water, surrounding itself with an 8- or 12-sided gold crystal layer.

Watterson and his colleagues, J.C. Antweiler and W. L. Campbell, wanted to know why certain remote river beds in Alaska were rich with gold nuggets while all the land upstream consisted of a gold-poor rock called chert.

Some of the nuggets weighed as much as one ounce, while the rocks upstream contained gold only in milligram-sized flakes.

Watterson told scientists at the recent International Symposium on Environmental Biochemistry in Santa Fe, N.M., that he took a weak solution of gold dissolved in water, added some bacillus cereus, and waited 36 hours. When he examined the tiny bacteria, he found each one had coated itself with a thin layer of gold crystal. Some crystals already were 15 times the diameter of the original spore, which was still trapped inside.

Once the crystal is formed in a solution, Watterson said, it continues to grow indefinitely, long after the original bacteria has died.

Watterson and his colleagues are sure these "bug nuggets" formed in the river gravel rather than washing down from rich gold veins because the nuggets are almost perfectly undisturbed gold crystals. If the gold washed down from an upstream vein, they said, the soft metal would have a rounded, polished look. But gold flakes and nuggets found in placer deposits worldwide have flat, unscratched crystal faces.

"They must have been formed right there, in the gravel," said Watterson. "Natural gold is very soft and registers abrasion if it is moved any distance.

"For years, the miners have been saying placer gold had to come from somewhere other than upstream.

"It's been a common observation that old placers reconstitute themselves. Take an old placer, mine it, wait 20 or 30 years, and a new bed of gold will have grown."

Though he hasn't yet examined placer gold from California's Mother Lode, results from Alaska, Brazil and British Columbia all confirm Watterson's discovery: When a gold nugget or crystal is cut open, remnants of bacteria are found inside.

'Take an old placer, mine it, wait 20 or 30 years, and a new bed of gold will have grown'

Because California has the rich, gold-bearing Mother Lode, Watterson speculated the placer gold in rivers here is a combination of bacterial crystallization and natural erosion.

"I haven't yet looked at California gold," he said, "but I would be happy to study (it) if someone would send some to me."

Watterson said biology works in at least two ways before gold flakes and nuggets can form.

First, gold has to be dissolved in river water. Many rocks contain trace amounts of gold, but an acid usually is required to leach it out. Nature provides that acid. Fungi and microorganisms living in the soil are known to produce acids that release soil minerals. Also, humic acid is produced by decaying plant matter. It is even possible that in the past few decades acid rain has contributed to the leaching process.

As gold-laden water cascades down rivers and streams, a number of bacteria use it in a never-ending struggle against one another.

Gold, Watterson said, is something of an antibiotic.

If a bacteria can find a way to become tolerant to the metal, it may then accumulate the metal to use as a poison against other, non-tolerant bacteria it competes against.

Watterson speculated that bacillus cereus has learned to build its own "time capsule." When covered with gold, the bacteria cannot germinate until something eventually dissolves away the gold. That, according to Watterson, "could take a long, long time."

Could the bacillus cereus have evolved this mechanism to protect its species against drought, predation or disease? No firm scientific evidence exists to prove the point, but bacillus cereus is remarkably hardy. It is found both in the arctic permafrost and the tropical Amazon River basin.

"For a long time, gold was thought to be inert," Watterson said, "but it is extremely biologically active and has some interesting properties."

He discovered some of gold's properties entirely by accident while measuring the "inhibition zones" — where bacteria will not grow — around various metals.

"We were all set up to do these tests and a guy I was working with said, 'Want to use gold too?' We threw it in just for the hell of it and we noticed these huge inhibition zones around gold."

Gold was acting like a bacterial contraceptive, he said.

Watterson did the inhibition zone experiments on his own time, for the sheer joy of it. "It's just something I do for fun," he said.

Fun or not, bacterial processes can be used to extract metals for profit too. About 20 percent of U.S. copper is extracted using a bacterial process to absorb low-grade ores.

"The Russians have set up a fairly elaborate study group to look at microorganisms for mining gold," said Watterson. "But so far, we aren't that imaginative.

BACILLUS CEREUS

John Watterson
J.C. Antweiler
W.L. Campbell